

Telliex Chiu

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WORK EXPERIENCE

Senior Full-Stack Software Engineer | Health Innovators | Cambridge, MA

Jan 2025 - Present

- Multilingual Healthcare Chatbot: Developed a healthcare platform using **React.js/Next.js** and **Vercel AI SDK** to support multilingual chatbot interactions for symptoms and drug inquiries. Integrated **Retrieval-Augmented Generation (RAG)** and a medical knowledge base to enable global users to interact via text and speech, including FDA drug recall reporting. Projected to improve query resolution rate by 20% and reduce adverse drug reporting time.

Senior Full-Stack Software Engineer | Awkns | Taipei, Taiwan

May 2024 - Apr 2025

- Designed a point redemption system for large-scale on-site events, featuring a reward mechanism and role-based point flow using **React.js/Next.js** and **Firestore**. Implemented **IndexedDB** for offline data access and secure sync under poor network conditions. Developed one-time QR code verification to prevent abuse. Integrated indoor navigation and live admin monitoring. Supported hundreds of stable transactions with error rates reduced to 2% and user engagement increased by 35%.
- Developed a **Stripe-integrated** subscription landing page for a SaaS product, supporting trials, upgrades, downgrades, renewals, and refunds. Designed scalable logic and access control to support multi-product expansion.
- Designed a high-concurrency online ticketing system using **React.js/Next.js** for the frontend. Integrated **Firestore Transactions**, **Stripe**, and one-time QR code validation to achieve real-time inventory synchronization and anti-fraud verification. Implemented **WR/SSE (Web Response / Server-Sent Events)** on the backend for real-time inventory and order status updates. Introduced **NFT-based** ticket validation to ensure ticket uniqueness and prevent forgery. Successfully reduced overselling errors to 0% and improved ticket verification speed to under 1 second.

Senior Full-Stack Software Engineer - Remote | ECloudyValley | Taipei, Taiwan

Apr 2022 - Apr 2024

- Led the development of an enterprise-grade financial reporting platform using **React.js** and **TypeScript**, integrating AWS billing and rebilling modules. The system supports multi-department, multi-role field-level customization and approval workflows, enabling fine-grained access control and dynamic report generation. Implemented asynchronous report generation and real-time notifications, reducing average report download time by 60% and manual effort by over 70%.
- Redesigned the financial processing workflow and caching strategy using **Python** batch jobs to automatically calculate and consolidate cloud billing data daily, significantly improving data integration efficiency and query performance. Built the backend with **Nest.js** to provide **RESTful APIs**, implementing field-level permission filtering and operation logging for comprehensive sensitive data access control and auditing.
- Developed an internal shared component library and standardized frontend state and data flow architecture to enhance development consistency and team collaboration. Introduced a modular architecture, **CI/CD pipelines**, and **Docker-based** automated deployment, ensuring platform stability and scalability for simultaneous access by multiple business units. The system has become a core tool for operational decision-making across the organization.

Senior Frontend Engineer & Frontend Leader | Shinho (ArcheX, Bits x Bites) | Shanghai, China

Dec 2015 - Mar 2022

- Enterprise Digital Management Platform for Manufacturing Plants
 - Led digital transformation initiatives across multiple factories by integrating **IoT** devices and production line data. Built real-time dashboards and operational platforms, successfully unifying data from all sites, improving real-time monitoring and data accuracy, and reducing operational and management costs by over 20%.
 - Drove the transition to a frontend microservices architecture using **React.js** and **TypeScript**, implementing modular design via **Module Federation** and **Nx Monorepo**. Enabled scalable, independently deployable modules supporting multi-site operations.
 - Introduced a mobile data entry app with **Flutter**, empowering frontline workers to log production parameters and anomalies in real time. Integrated with **WebSocket APIs** to display live temperature, humidity, machine status, and production metrics, enhancing on-site and backend decision-making efficiency.
- Mobile Ordering via QR Code and Self-Service Kiosks for F&B Chains
 - Architected and developed the frontend for mobile **QR code** ordering and self-service kiosks, delivering an intuitive, responsive UI for multi-device use. Improved ordering speed and customer satisfaction.

- Implemented real-time menu synchronization and order processing via **WebSockets**, integrated with POS systems to automate kitchen prep and printing processes, reducing human error and latency. Optimized concurrency logic for group ordering to maintain system stability during peak hours.
- Integrated **Google Analytics** and custom event tracking to analyze user behavior and preferences, producing visual reports that helped optimize menus and promotions, resulting in a 20% increase in top-selling items and stronger revenue growth.
- Online Dessert Course Purchasing and Offline Attendance System
 - Developed a full-featured course purchasing platform using **React.js** and **Redux**, covering product display, bundled course selection, inventory management, and real-time price updates. Used modular development and performance tuning (**Lazy Load, Virtual DOM optimization**) to improve load time by 30%.
 - Designed a flexible transaction and promotion engine supporting complex rules, including course credits, points redemption, tiered discounts, and time-limited offers. Integrated shopping cart validation and checkout error handling, reducing transaction failures by 20% and increasing overall success rate and user satisfaction.
 - Built user behavior analytics and conversion funnel insights via **Google Analytics** and in-house tracking tools, optimizing key courses and marketing flows. Collaborated with marketing to boost specific course sales by 10% and improved user retention and repeat purchases.
- Group Brand Website and Backend Management System
 - Redesigned corporate brand websites using **Vue.js** to improve visibility and UX. Unified multiple brand identities and optimized **SEO** through **SSR (Server-Side Rendering)**, **responsive design (RWD)**, and structured data, achieving a 15% boost in search ranking and consistent cross-device performance.
 - Built a **Node**-based CMS for dynamic content and multi-user permission management. Designed **RESTful APIs** integrated with **MySQL**, optimized database queries, and deployed caching/CDN strategies to cut page load times by 20%.
- F&B Visual Content Scheduling and Playback System
 - Developed a cross-store visual content scheduling platform using **React.js** and **Electron** for in-store playback, supporting screen tiling and content synchronization. Backend built with **Node.js** and **WebSockets** for real-time updates and error handling, improving playback stability and automation.
 - Designed a modular CMS for content scheduling and device monitoring, including scheduling templates and sensor **APIs** for future interactive capabilities. The system was deployed across all Shanghai stores and applied in co-branded campaigns with other brands. Weekly update frequency increased from monthly to become a core part of the smart retail infrastructure.

Frontend Creative Engineer | Yahoo | Taipei, Taiwan

May 2010 - Nov 2015

- Designed and developed cross-platform rich media ad products using **HTML5**, **JavaScript**, and **CSS/CSS3**, delivering 4+ new products annually. Improved compatibility and UX, driving 10% yearly revenue growth through increased client engagement.
- Developed a **Python**-based system to automate real-time ad performance tracking via **RESTful APIs**. Enabled **A/B testing** and dynamic content adjustment, boosting CTR by 15% and reducing feedback latency by 80%. Collaborated with marketing and frontend teams to improve campaign optimization and decision-making.
- Led the transition from Flash to modular **JavaScript + HTML5 + CSS3** ad components, supporting multiple formats and brand customization. Parameterized design enhanced reusability and maintainability, cutting development time by ~30% and improving engagement by 15%.

PROJECTS

Activo.quest | Personal Portfolio & AI Experimentation | San Jose, CA | 2024 - Present

- Developed a multilingual voice assistant with **Next.js**, **Grok API**, **Llama3**, and **Cartesia TTS**, integrating domain-specific knowledge bases for intelligent Q&A in English and Mandarin.
- Built a **Deepgram**-powered web app enabling real-time English and Mandarin speech transcription and voice commands, supporting multi-speaker dialogue recording with an intuitive interface.

SKILLS

- Languages: **React.js**, **Next.js**, **Vue.js**, **TypeScript**, **Tailwind**, **Node.js**, **Nest.js**, **Python**, **Zustand**
- Database: **MongoDB**, **MySQL**, **PostgreSQL**, **Firebase**

- Tools: AWS, Docker, Vercel, Vercel AI SDK
- Certifications: AWS Solutions Architect, AWS DevOps Engineer, PMI PMP, ScrumMaster

EDUCATION

B.S. in Biology | Chinese Culture University, Taipei, Taiwan